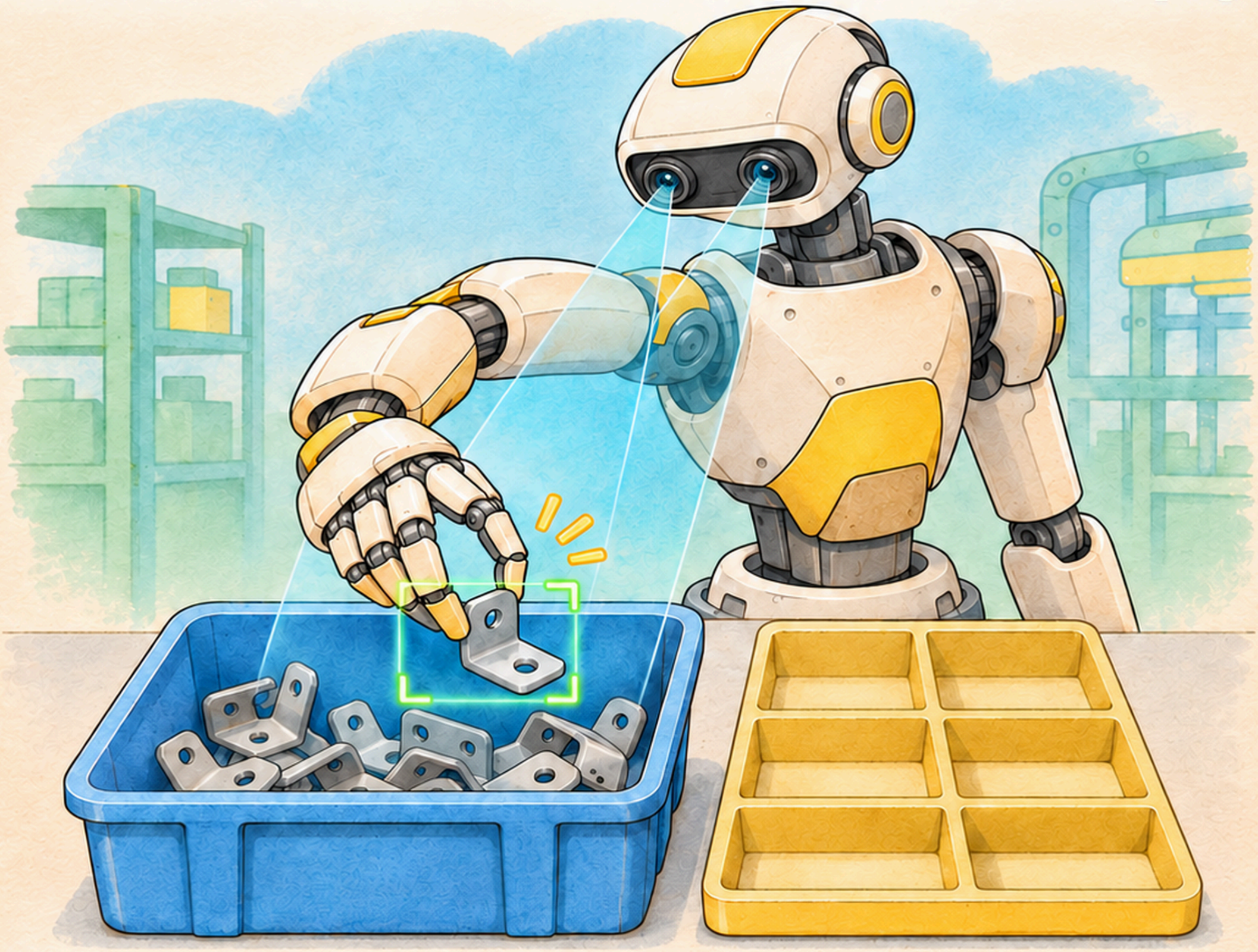
How does AI help a robot pick up a part?

Gemini Robotics 2.5, explained simply

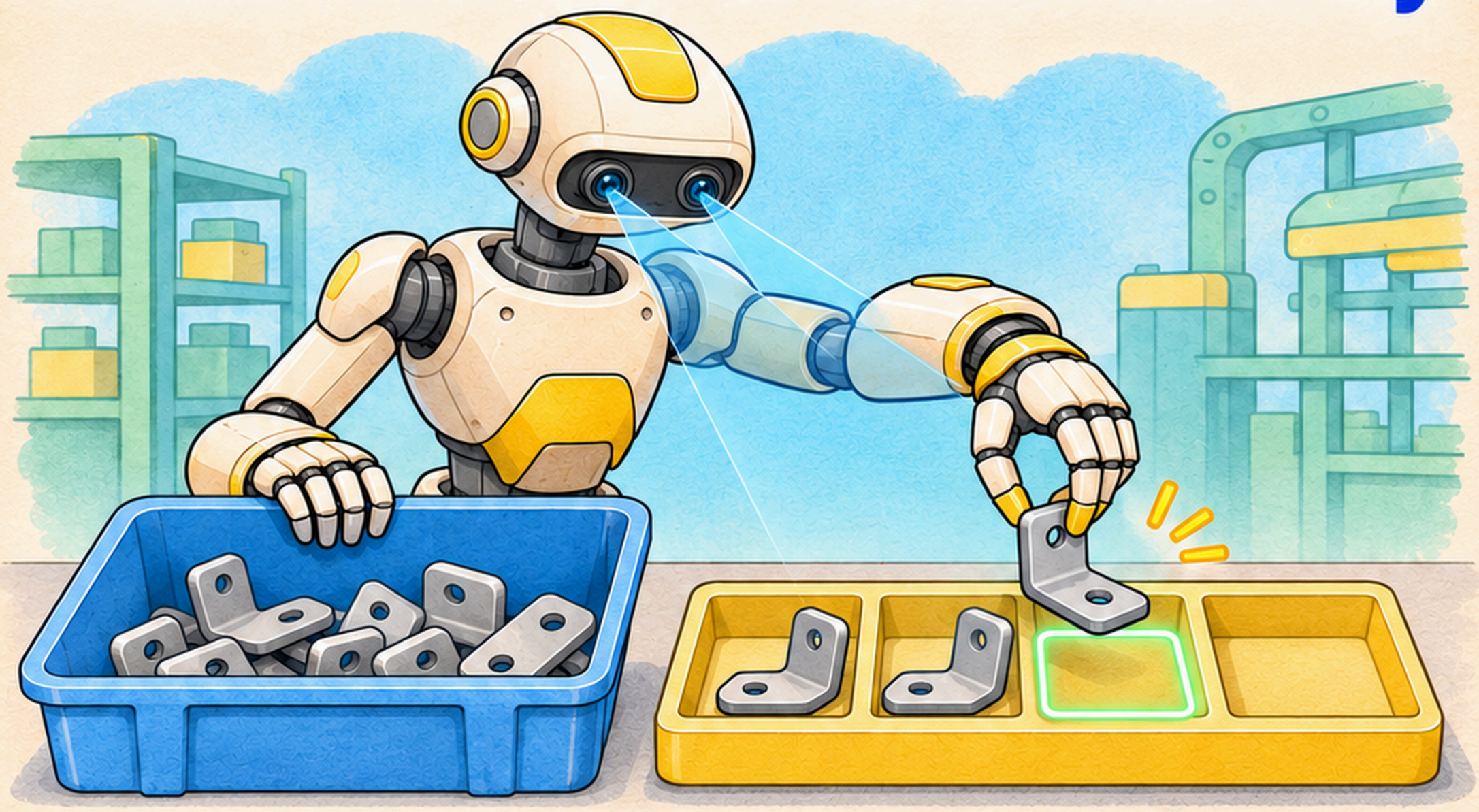


A camera, a task, and a moving hand.

Illustration; Google-reported pilot

A real factory task

From a bin to an ordered tray



The robot picks metal brackets from a bin and places them in a tray in the order needed.

Google reports this was a factory pilot with Apptronik and Mercedes-Benz.

Illustration, not a pilot photo

What does the model need?

Two inputs give the robot context

VISION

Camera images show the parts and their surroundings.

LANGUAGE

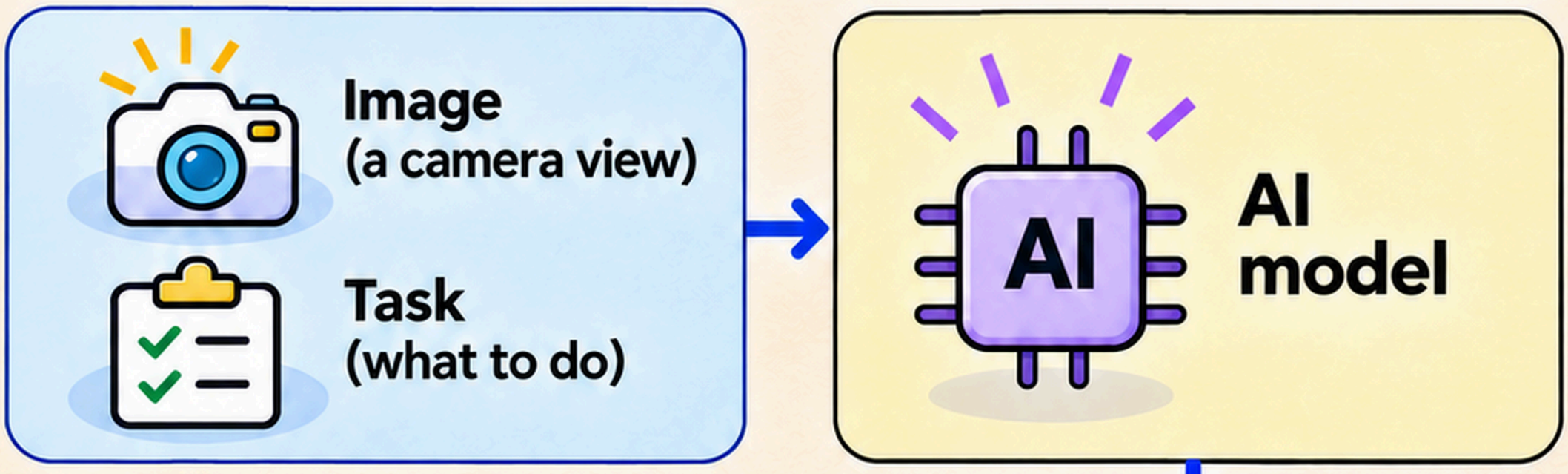
A task instruction describes what the robot needs to do.

ROBOTICS MODEL

The model uses both inputs to select actions.

From image to movement

The AI model produces robot actions



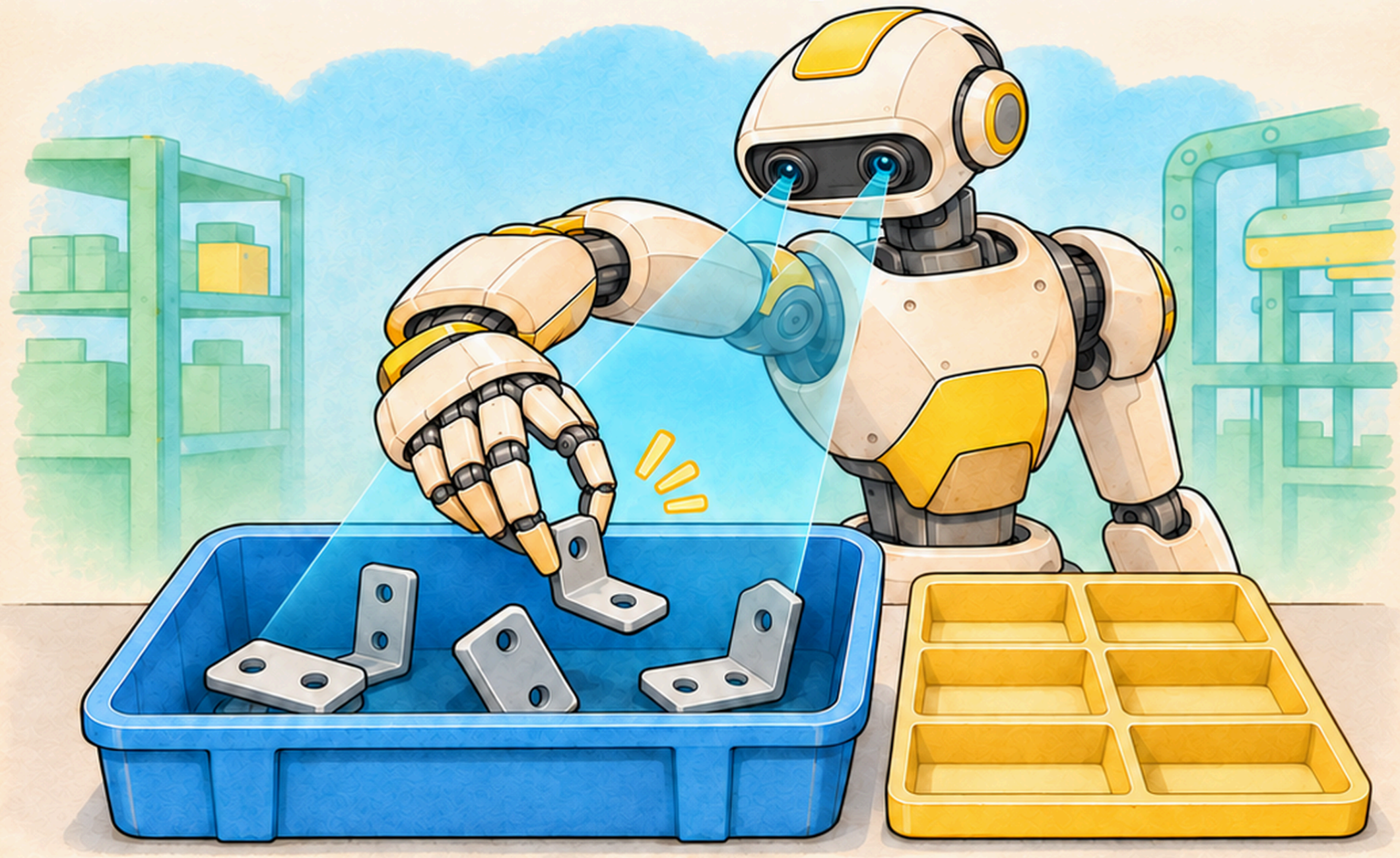
The model uses the image and task to produce robot actions.

Those actions guide the arm and hand to reach, grip and place.

A simplified view of the system.

Why is this difficult?

The robot must adapt to the scene



Parts can overlap, shift, or face different directions.

The robot must adjust its movements as the scene changes.

Google reports the robot adjusted while sorting parts.

What did Google report?

A factory pilot, not a universal promise

Apollo 2 sorted parts in a Mercedes-Benz factory pilot.

One model tracked the workflow; Gemini Robotics 2.5 guided the hands and arms.

One pilot does not prove performance everywhere.

What task would you teach a robot?

Source: Google DeepMind